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EXPERIMENT 10

Performance optimization

Aim

Measure and improve library report and workflow performance through selective queries, sensible relationships and bounded processing.

Procedure

1. Create a synthetic development dataset within the account limits, for example 100 titles, 300 copies, 200 members and 2,000 loan-history rows. Record the actual sizes used and preserve a reproducible fixture generator or import file.

2. Measure the same dashboard, overdue report and member lookup repeatedly with the same role, device and dataset. Record a warm-up separately, then at least ten timings. Report median and range; do not invent a speedup.

3. Replace full-table loops with criteria-based queries and aggregates. Fetch each required record once per operation. Avoid a remote API call inside a loop when batching or a queued process is available.

4. Limit initial report rows, visible columns and date range; use API pagination for bulk retrieval. Split heavy dashboard content into focused pages and load only the components needed for the user’s task.

5. Keep Books and Members as standalone forms linked by lookups. Do not embed all a member’s loan history inside an ever-growing subform. Use subforms only for bounded child rows, such as a small acquisition batch, with an explicit row limit and measured performance.

Implementation and verification

6. Discuss indexing as a database concept, then inspect the actual Creator controls and documentation. Do not issue SQL CREATE INDEX or claim access to arbitrary database indexes. Record any supported field-index option if present; otherwise use platform-managed indexing and document the limitation.

7. Apply one change at a time and repeat measurements. Confirm report totals, role restrictions and loan rules still match the baseline.

```
// Select only relevant records rather than scanning every loan.
overdue = Loans[Status == "Issued" && Due_Date < zoho.currentdate];
openCount = Loans[Member == memberId && Status == "Issued"].count();
```

Operation	Before median	After median	Correctness
Overdue report	_____ ms	_____ ms	Same IDs and count
Dashboard	_____ ms	_____ ms	Same panels
Member open count	_____ ms	_____ ms	Same result

No Duplicate values is a data-integrity setting, not proof of a measured indexing improvement. For historical archiving, preserve identifiers and auditability and confirm which reports intentionally exclude archived data.

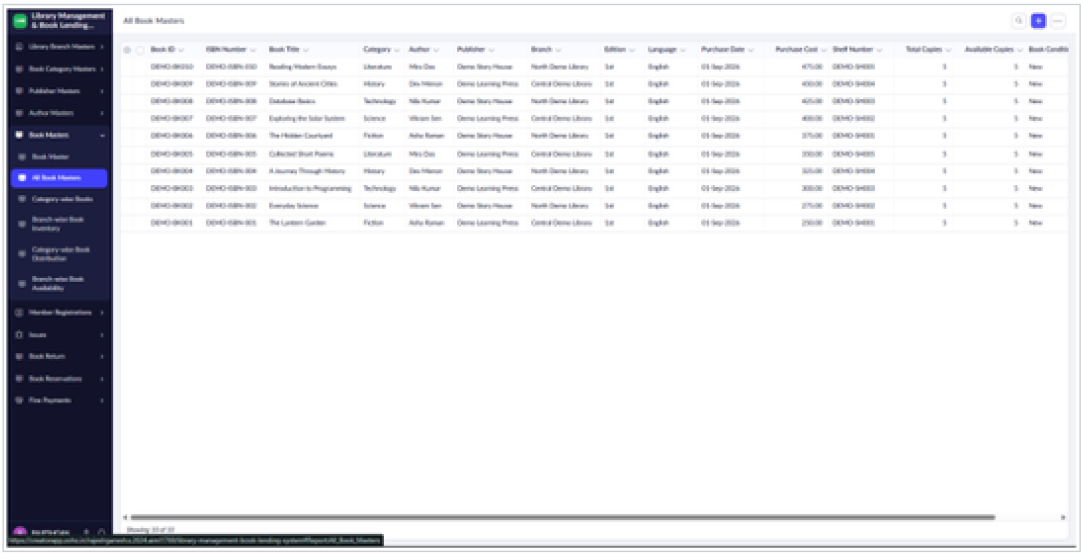
Implementation and interpretation

Use a fixed dataset for comparisons and preserve the same access filters. A faster report that omits required records is not a successful optimization. Record timing distributions rather than presenting a single unusually fast response as the result.

EXPERIMENT 10 VISUAL RECORD

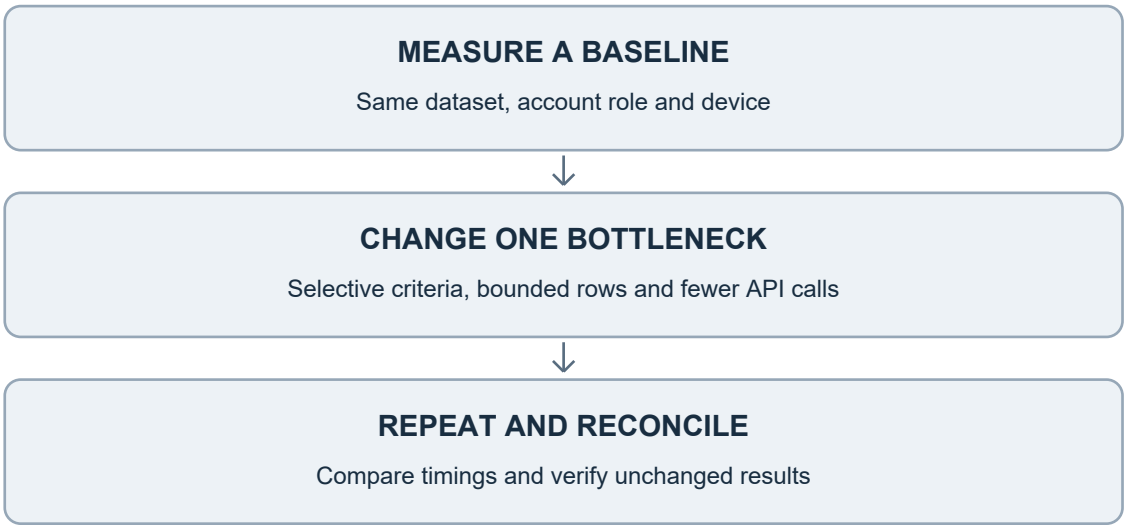
Application images and explanation

Book Masters report used as a performance test surface



Supplied application screenshot from the uploaded library document.

Illustrative process diagram



Generated explanatory diagram. It is not a screenshot of an executed workflow.

Use a fixed dataset for comparisons and preserve the same access filters. A faster report that omits required records is not a successful optimization. Record timing distributions rather than presenting a single unusually fast response as the result.

Result

Expected result and verification criteria. Measured improvements are reported with dataset and test conditions, and functional results remain unchanged. Unsupported index control is documented honestly; no performance guarantee is inferred from a small dataset.